

Mustafa Jarrar: Lecture Notes in Discrete Mathematics.
Birzeit University, Palestine, 2021

Set Theory



6.1. Basics of Set Theory

6.2 Properties of Sets

6.3 Algebraic Proofs

6.4 Boolean Algebras



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<http://jarrar-courses.blogspot.com/2014/03/discrete-mathematics-course.html>

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Acknowledgement:

This lecture is based on (but not limited to) to chapter 5 in "Discrete Mathematics with Applications by Susanna S. Epp (3rd Edition)".

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Set Theory

6.1 Basics of Sets

In this lecture:

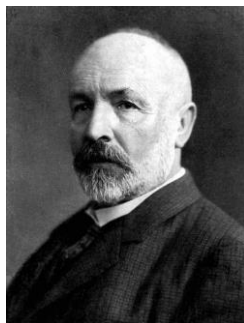


☒ Part 1: Basic Concepts and Notations

- ☐ Part 2: Subsets, proper subsets, and Set Equalities
- ☐ Part 3: Operations on Sets
- ☐ Part 4: Empty Sets
- ☐ Part 5: Partitions of Sets
- ☐ Part 6: Power Sets & Cartesian Products

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History



Georg Cantor

1845 – 1918

Born in Saint Petersburg, Russia

Moved to Germany 1856

PhD: University of Berlin 1867

Work: University of Halle

Set theory is the branch of mathematical logic that studies sets, which informally are collections of objects.

Initiated by Georg Cantor in 1870s

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Basic Concepts and Notations

Cantor suggested a set as a:

“collection into a whole ***M*** of definite and separate objects of our intuition or our thought”.

$$\mathbf{M} = \{ \text{Ali, Adam, Sara} \}$$

Each object is called an elements (or member of) of ***M***.

$\text{Ali} \in \mathbf{M}$ (Ali belongs to *M*)

$\text{Rami} \notin \mathbf{M}$ (Rami does not belong to *M*)

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Basic Concepts and Notations

The order of elements is irrelevant

$$\{\text{Ali, Adam, Sara}\} = \{\text{Adam, Sara, Ali}\}$$

Redundancy is not allowed

$$\{\text{Ali, Adam, Adam, Sara}\}$$

A set can be an element inside another set

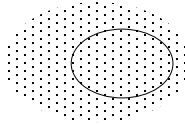
$$\{1, \{1\}\} \quad \text{has two elements}$$

Notation of elements

$$\{\text{Ali}\} \neq \text{Ali} \quad \text{different elements}$$

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Defining Sets by a Property



$$A = \{x \in S \mid P(x)\}$$

The set of all
"x is dummy"

Property

Examples:

The set of all integers that are more than -2 and less than 5

$$\{x \in \mathbb{Z} \mid -2 < x < 5\}$$

The set of all persons who born in Palestine

$$\{x \in \text{Person} \mid \text{BornIn}(x, \text{Palestine})\}$$

The set of all persons who born in Palestine and Love Homus

$$\{x \in \text{Person} \mid \text{BornIn}(x, \text{Palestine}) \wedge \text{Love}(x, \text{Homus})\}$$

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Set Versus Element

In Set theory → **Set vs. Element** ← Mathematical Set

In JAVA → Class vs. Object

- The **extension** of a set is its elements.
- The **order** of elements is irrelevant
- In set theory: an element itself might be a set.

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Set Theory

6.1 Basics of Sets

In this lecture:

☐ Part 1: Basic Concepts and Notations

 ☒ **Part 2: Subsets, proper subsets, and Set Equalities**

☐ Part 3: Operations on Sets

☐ Part 4: Empty Sets

☐ Part 5: Partitions of Sets

☐ Part 6: Power Sets & Cartesian Products

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Subsets

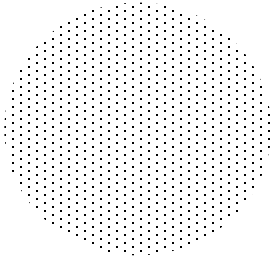
المجموعة الجزئية

$$A \subseteq B \Leftrightarrow \forall x, \text{ if } x \in A \text{ then } x \in B.$$



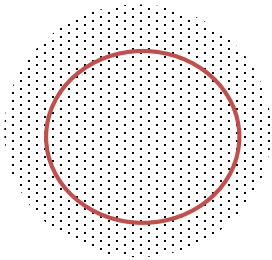
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Subsets Versus JAVA SubClasses



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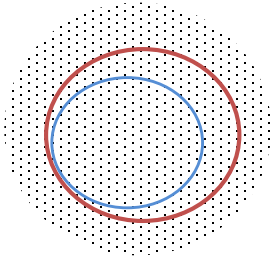
Subsets Versus JAVA SubClasses



Animal = { $x \in \text{LivingOrganism} \mid \text{CanMove}(x)$ }

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Subsets Versus JAVA SubClasses



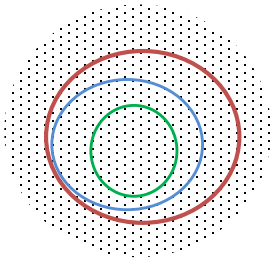
Animal = $\{x \in \text{LivingOrganism} \mid \text{CanMove}(x)\}$

Human = $\{x \in \text{Animal} \mid \text{HasMind}(x) \wedge \text{Utter}(x)\}$

Human $\subset$ Animal

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Subsets Versus JAVA SubClasses



Animal = $\{x \in \text{LivingOrganism} \mid \text{CanMove}(x)\}$

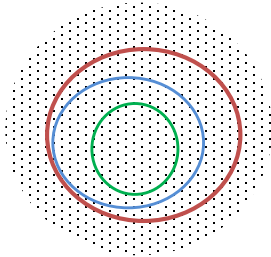
Human = $\{x \in \text{Animal} \mid \text{HasMind}(x) \wedge \text{Utter}(x)\}$

Woman = $\{x \in \text{Human} \mid \text{GivesBirth}(x)\}$

Woman $\subset$ Human $\subset$ Animal

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Subsets Versus JAVA SubClasses

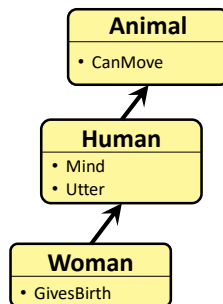


Animal = $\{x \in \text{LivingOrganism} \mid \text{CanMove}(x)\}$

Human = $\{x \in \text{Animal} \mid \text{HasMind}(x) \wedge \text{Utter}(x)\}$

Woman = $\{x \in \text{Human} \mid \text{GivesBirth}(x)\}$

Woman $\subset$ Human $\subset$ Animal



Every subclass inherits the properties of its super class, thus:

- Human is a living organism that can move, has mind and utter.
- Woman is a living organism that can move, has mind and utter, and able to give birth.

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Distinction between $\in$ and $\supseteq$

Which of the following are true statements?

$$2 \in \{1, 2, 3\}$$

$$\{2\} \in \{1, 2, 3\}$$

$$2 \subseteq \{1, 2, 3\}$$

$$\{2\} \subseteq \{1, 2, 3\}$$

$$\{2\} \subseteq \{\{1\}, \{2\}\}$$

$$\{2\} \in \{\{1\}, \{2\}\}$$

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Distinction between $\in$ and $\supseteq$

Which of the following are true statements?

✓ $2 \in \{1, 2, 3\}$

✗ $\{2\} \in \{1, 2, 3\}$

✗ $2 \subseteq \{1, 2, 3\}$

✓ $\{2\} \subseteq \{1, 2, 3\}$

✗ $\{2\} \subseteq \{\{1\}, \{2\}\}$

✓ $\{2\} \in \{\{1\}, \{2\}\}$

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Subsets Notations

Not Subset:

$$A \not\subseteq B \Leftrightarrow \exists x . x \in A \text{ and } x \notin B$$

Notations:

$$A = B$$

A equals B

$$A \subset B$$

$$B \supset A$$

A is subset of B

$$A \subseteq B$$

$$B \supseteq A$$

A is subset or equal of B

$$A \not\subset B$$

$$B \not\supset A$$

A is not a subset of B

$$A \not\subseteq B$$

$$B \not\supseteq A$$

A is not a subset or equal of B

$$A \subsetneq B$$

$$B \supsetneq A$$

A is a subset but not equal of B

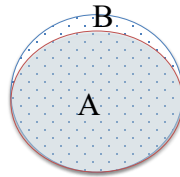
Examples: $\text{Person} \supseteq \text{Man}$, $\mathbb{Z} \supseteq \mathbb{Z}^+$, $\mathbb{R} \not\subset \mathbb{Z}$

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Proper Subsets

Definition

Let A and B be sets. A is a **proper subset** of B if, and only if, every element of A is in B but there is at least one element of B that is not in A.



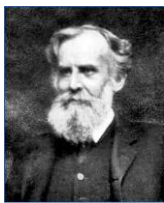
Examples of proper subsets:

$$\{-2, -3, 4\} \subset \{-2, -2.5, -3, -3.5, -4\}$$

$$\{1, 2, 3, 4, \dots\} \subset \{0, 1, 2, 3, 4, \dots\}$$

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Venn Diagrams

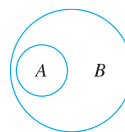


John Venn, British
(1834-1923)

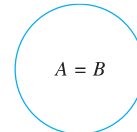
Represented sets as diagrams in 1881.
used to teach elementary set theory,



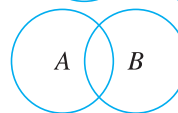
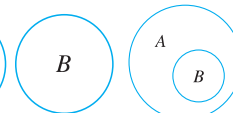
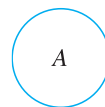
$$A \subseteq B$$



$$A = B$$



$$A \not\subseteq B$$



Z: integers (صحبة)

Q: rational numbers (نسبية)

R: real numbers (حقيقية)



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Proving and Disproving Subset Relations

Define sets A and B as follows:

$$A = \{m \in \mathbb{Z} \mid m = 6r + 12 \text{ for some } r \in \mathbb{Z}\}$$

$$B = \{n \in \mathbb{Z} \mid n = 3s \text{ for some } s \in \mathbb{Z}\}.$$

Prove that $A \subseteq B$.

Suppose x is a particular but arbitrarily chosen element of A .

Show that $x \in B$, means show that $x = 3 \cdot (\text{integer})$.

$$\begin{aligned} x &= 6r + 12 \\ &= 3 \cdot (2r + 4). \end{aligned}$$

$$\text{Let } s = 2r + 4.$$

$$\begin{aligned} \text{Also, } 3s &= 3(2r + 4) \\ &= 6r + 12 \\ &= x \end{aligned}$$

Therefore, x is an element of B .

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Set Equality

Definition

Given sets A and B , A **equals** B , written $A = B$, if, and only if, every element of A is in B and every element of B is in A .

Symbolically: $A=B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A$.

Example: Define sets A and B as follows:

$$A = \{m \in \mathbb{Z} \mid m = 2a \text{ for some integer } a\}$$

$$B = \{n \in \mathbb{Z} \mid n = 2b - 2 \text{ for some integer } b\}$$

Is $A = B$?

Yes. To prove this, both subset relations $A \subseteq B$ and $B \subseteq A$ must be proved.

Part 1, Proof That $A \subseteq B$:

Suppose x is a particular but arbitrarily chosen element of A .
[We must show that $x \in B$. By definition of B , this means we must show that $x = 2 \cdot (\text{some integer}) - 2$.]

By definition of A , there is an integer a such that $x = 2a$.
[Given that $x = 2a$, can x also be expressed as $2 \cdot (\text{some integer}) - 2$? i.e., is there an integer, say b , such that $2a = 2b - 2$?
Solve for b to obtain $b = (2a + 2)/2 = a + 1$. Check to see if this works.]

Part 1, Proof That $A \subseteq B$:

Part 2, Proof That $B \subseteq A$:

Part 2, Proof That $B \subseteq A$:

Suppose y is a particular but arbitrarily chosen element of B .
[We must show that $y \in A$. By definition of A , this means we must show that $y = 2 \cdot (\text{some integer})$.]

By definition of B , there is an integer b such that $y = 2b - 2$.
[Given that $y = 2b - 2$, we want to see if y can be written in the form $2a$. So we set $2a = 2b - 2$ and solve for a : $a = (2b - 2)/2 = b - 1$.]

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Let $b = a + 1$.
[First check that b is an integer.]

Then b is an integer because it is a sum of integers.
[Then check that $x = 2b - 2$.]

Also $2b - 2 = 2(a + 1) - 2 = 2a + 2 - 2 = 2a = x$.
Thus, by definition of B , x is an element of B
[which is what was to be shown].

Let $a = b - 1$.
[First check that a is an integer.]

Then a is an integer because it is a difference of integers.
[Then check that $y = 2a$.]

Also $2a = 2(b - 1) = 2b - 2 = y$.
Thus, by definition of A , y is an element of A .
[which is what was to be shown].

Set Theory

6.1 Basics of Sets

In this lecture:

- ☐ Part 1: Basic Concepts and Notations
- ☐ Part 1: Subsets, proper subsets, and Set Equalities
-  ☒ Part 3: **Set Operations** (Union, Intersection, Difference, Complement)
- ☐ Part 4: Empty Sets
- ☐ Part 5: Partitions of Sets
- ☐ Part 6: Power Sets & Cartesian Products

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Operations on Sets

• Definition

Let A and B be subsets of a universal set U .

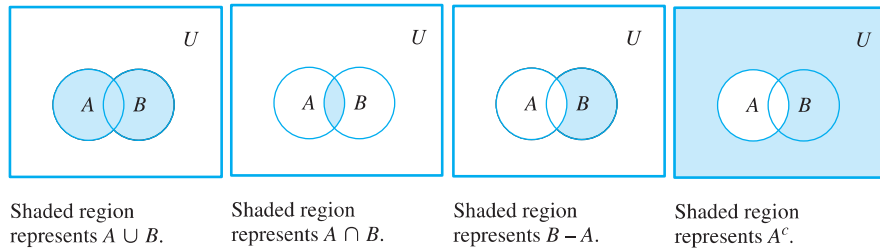
1. The **union** of A and B , denoted $A \cup B$, is the set of all elements that are in at least one of A or B .
2. The **intersection** of A and B , denoted $A \cap B$, is the set of all elements that are common to both A and B .
3. The **difference** of B minus A (or **relative complement** of A in B), denoted $B - A$, is the set of all elements that are in B and not A .
4. The **complement** of A , denoted A^c , is the set of all elements in U that are not in A .

Symbolically:

$$\begin{aligned}
 A \cup B &= \{x \in U \mid x \in A \text{ or } x \in B\}, \\
 A \cap B &= \{x \in U \mid x \in A \text{ and } x \in B\}, \\
 B - A &= \{x \in U \mid x \in B \text{ and } x \notin A\}, \\
 A^c &= \{x \in U \mid x \notin A\}.
 \end{aligned}$$

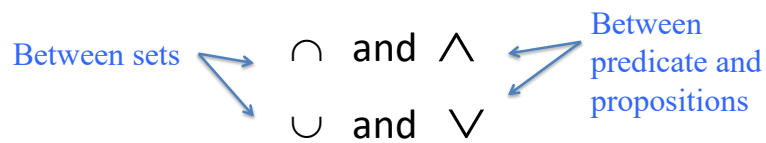
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Operations on Sets



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Distinction between $\cap$ and $\wedge$



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Indexed Collection of Sets

• Definition

Unions and Intersections of an Indexed Collection of Sets

Given sets $A_0, A_1, A_2, \dots$ that are subsets of a universal set U and given a nonnegative integer n ,

$$\bigcup_{i=0}^n A_i = \{x \in U \mid x \in A_i \text{ for at least one } i = 0, 1, 2, \dots, n\}$$

$$\bigcup_{i=0}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for at least one nonnegative integer } i\}$$

$$\bigcap_{i=0}^n A_i = \{x \in U \mid x \in A_i \text{ for all } i = 0, 1, 2, \dots, n\}$$

$$\bigcap_{i=0}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for all nonnegative integers } i\}.$$

Non-negative: Tells us where the collection of sets starts (starts from zero: 0, 1, 2, ... and continues to infinity).
For all: Tells us the condition for the intersection (that the element must exist in all sets, whether the total number of sets is n or infinity).
At least: Tells us the condition for the union (that it is enough for the element to exist in at least one set, whether the total number of sets is n or infinity).

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Finding Unions and Intersections of More than Two Sets

For each positive integer i , let $A_i = \left\{x \in \mathbb{R} \mid -\frac{1}{i} < x < \frac{1}{i}\right\} = A_i = \left(-\frac{1}{i}, \frac{1}{i}\right)$

A_1 : set of all real numbers between -1 and 1

A_2 : set of all real numbers between -1/2 and 1/2

A_3 : set of all real numbers between -1/3 and 1/3

Find $A_1 \cup A_2 \cup A_3 = (-1, 1)$, because $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $\left(-\frac{1}{3}, \frac{1}{3}\right)$ are included

Find $A_1 \cap A_2 \cap A_3 = \left(-\frac{1}{3}, \frac{1}{3}\right)$, because $(-1, 1)$ and $\left(-\frac{1}{2}, \frac{1}{2}\right)$ are included

Find $\bigcup_{i=1}^{\infty} A_i = (-1, 1)$ Find $\bigcap_{i=1}^{\infty} A_i = \emptyset$

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Formalizing Statements in Set Theory

- All smart students

$$\text{Smart} \cap \text{Student}$$

- Students who are not Smart

$$\text{Student} \cap \text{Smart}^c \text{ or } \text{Student} - \text{Smart}$$

- There are no smart students from Palestine

$$\text{Smart} \cap \text{Student} \cap \text{Palestinian} = \emptyset$$

- There are no smart students from Palestine among the winners

$$\text{Smart} \cap \text{Student} \cap \text{Winner} \cap \text{Palestinian} = \emptyset$$

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Set Theory

6.1 Basics of Sets

In this lecture:

- ☐ Part 1: Basic Concepts and Notations
- ☐ Part 1: Subsets, proper subsets, and Set Equalities
- ☐ Part 3: Operations on Sets
-  ☐ Part 4: **Empty Sets**
- ☐ Part 5: Partitions of Sets
- ☐ Part 6: Power Sets & Cartesian Products

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The Empty Set

The empty set is not the same thing as nothing; rather, it is a set with nothing inside it and a set is always something. This issue can be overcome by **viewing a set as a bag—an empty bag undoubtedly still exists.**

Describe the set $D = \{x \in \mathbf{R} \mid 3 < x < 2\}$.

$$\forall A: \emptyset \subseteq A$$

$$\forall A: A \cup \emptyset$$

$$\subseteq A$$

$$\forall A: A \cap \emptyset$$

$$\subseteq \emptyset$$

$$\forall A: A \times \cup \emptyset$$

$$= \cup \emptyset$$

$$\forall A: A \times \cup \emptyset \Rightarrow A$$

$$= \cup \emptyset$$

While the empty set is a standard and widely accepted mathematical concept, it remains an ontological curiosity, **whose meaning and usefulness are debated by philosophers and logicians.**

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- ☐ Part 6: Power Sets & Cartesian Products

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Disjoint Sets

• Definition

Two sets are called **disjoint** if, and only if, they have no elements in common. Symbolically:

$$A \text{ and } B \text{ are disjoint} \Leftrightarrow A \cap B = \emptyset.$$

$$\text{Man} \cap \text{Woman} = \emptyset$$

• Definition

Sets $A_1, A_2, A_3, \dots$ are **mutually disjoint** (or **pairwise disjoint** or **nonoverlapping**) if, and only if, no two sets A_i and A_j with distinct subscripts have any elements in common. More precisely, for all $i, j = 1, 2, 3, \dots$

$$A_i \cap A_j = \emptyset \quad \text{whenever } i \neq j.$$

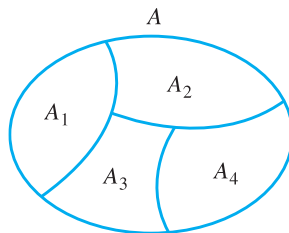
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Partitions of Sets

• Definition

A finite or infinite collection of nonempty sets $\{A_1, A_2, A_3, \dots\}$ is a **partition** of a set A if, and only if,

1. A is the union of all the A_i
2. The sets $A_1, A_2, A_3, \dots$ are mutually disjoint.



$$\begin{aligned} \text{Man} \cap \text{Woman} &= \emptyset \\ \text{Person} &= \text{Man} \cup \text{Woman} \end{aligned}$$

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Example

Let $\mathbf{Z}$ be the set of all integers and let

$$T_0 = \{n \in \mathbf{Z} \mid n = 3k, \text{ for some integer } k\},$$

$$T_1 = \{n \in \mathbf{Z} \mid n = 3k + 1, \text{ for some integer } k\},$$

$$T_2 = \{n \in \mathbf{Z} \mid n = 3k + 2, \text{ for some integer } k\}.$$

Is $\{T_0, T_1, T_2\}$ a partition of $\mathbf{Z}$?

Yes. By the quotient-remainder theorem, every integer n can be represented in exactly one of the three forms

$$n=3k \text{ or } n=3k+1 \text{ or } n=3k+2$$

It also implies that every integer is in one of the sets T_0 , T_1 , or T_2 .

$$\text{So } \mathbf{Z} = T_0 \cup T_1 \cup T_2.$$

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Power Sets

• Definition

Given a set A , the **power set** of A , denoted $\mathcal{P}(A)$, is the set of all subsets of A .

Find the power set of the set $\{x, y\}$. That is, find $\mathcal{P}(\{x, y\})$

$$= \{\emptyset, \{x\}, \{y\}, \{x, y\}\}.$$

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n -tuples

• Definition

Let n be a positive integer and let $x_1, x_2, \dots, x_n$ be (not necessarily distinct) elements. The **ordered n -tuple**, $(x_1, x_2, \dots, x_n)$, consists of $x_1, x_2, \dots, x_n$ together with the ordering: first x_1 , then x_2 , and so forth up to x_n . An ordered 2-tuple is called an **ordered pair**, and an ordered 3-tuple is called an **ordered triple**.

Two ordered n -tuples $(x_1, x_2, \dots, x_n)$ and $(y_1, y_2, \dots, y_n)$ are **equal** if, and only if, $x_1 = y_1, x_2 = y_2, \dots, x_n = y_n$.

Symbolically:

$$(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n) \Leftrightarrow x_1 = y_1, x_2 = y_2, \dots, x_n = y_n.$$

In particular,

$$(a, b) = (c, d) \Leftrightarrow a = c \text{ and } b = d.$$

Order n -tuples:

Is $(1, 2) = (2, 1)$? No. In ordered n -tuples, order matters.
For two n -tuples to be equal, their corresponding elements must be exactly the same.
Here, the first element 1 does not equal the second tuple's first element 2 ($1 \neq 2$).

Is $(3, (-2)^2, 1/3) = (\sqrt{9}, 4, \frac{3}{9})$? Since all corresponding elements are exactly equal, the two ordered triples are equal

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Cartesian Products

• Definition

Given sets $A_1, A_2, \dots, A_n$, the **Cartesian product** of $A_1, A_2, \dots, A_n$ denoted $A_1 \times A_2 \times \dots \times A_n$, is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ where $a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n$.

Symbolically:

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\}.$$

In particular,

$$A_1 \times A_2 = \{(a_1, a_2) \mid a_1 \in A_1 \text{ and } a_2 \in A_2\}$$

is the Cartesian product of A_1 and A_2 .

Example: Let $A_1 = \{x, y\}$, $A_2 = \{1, 2, 3\}$, and $A_3 = \{a, b\}$.

$$\begin{aligned} A_1 \times A_2 &= \\ &= \{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\} \end{aligned}$$

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Example

Let $A = \{\text{Ali, Ahmad}\},$
 $B = \{\text{AI, Dmath, DB}\},$
 $C = \{\text{Pass, Fail}\}$

Find $(A \times B) \times C =$

Find $A \times B \times C =$

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